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A comparison of the accuracy of Tzanakis and Alvarado Score in the diagnosis of acute appendicitis: A systematic review and meta-analysis

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ABSTRACT

Background and objectives: Acute appendicitis is one of the most commonly encountered surgical emergencies on a global level. Due to the requirement of an immediate clinical diagnosis and the presence of limited resources, clinicians and diagnosticians refer to scoring systems to diagnose this condition, among which Alvarado and Tzanakis scoring systems are widely used. This meta-analysis aims to compare the diagnostic accuracy of these two systems.

Methods: We searched PubMed, Google Scholar, and SCOPUS databases. All studies that reported diagnostic parameters of Alvarado and Tzanakis scores in patients with suspected acute appendicitis were selected. Diagnostic values such as sensitivity, specificity, positive predictive value (PPV), negative predictive value (NPV), and diagnostic accuracy were extracted from the selected studies and statistical analysis was performed with Meta Disc 1.4 software. Quality assessment of the selected studies was performed using the QUADAS-2 and QUADAS-C tools. Fourteen studies were included in our meta-analysis which enrolled 2235 patients.

Results: The overall sensitivity of the Tzanakis score was calculated as 0.86 (95% CI; 0.84-00.87) while the specificity was 0.73 (95% CI; 0.69–0.78). In addition, the area under the curve (AUC) was 0.9261 (SE; 0.0169) and the diagnostic Odds Ratio (OR) was 22.52 (95% CI; 9.47–53.56). The pooled sensitivity of Alvarado score was 0.67 (95% CI; 0.65–0.69) and the specificity was 0.74 (95% CI; 0.69–0.79). Moreover, the area under the curve (AUC) of the Alvarado score was 0.7389 (SE; 0.0489) and the diagnostic Odds Ratio was 4.92 (95% CI; 2.48–9.75).

Interpretation and conclusion: The Tzanakis scoring system has a higher sensitivity, area under the curve, and diagnostic odds ratio when compared to the Alvarado score. However, the Alvarado score has a marginally better specificity making it more reliable in excluding acute appendicitis.

Introduction

Acute appendicitis is the most common cause of abdominal surgical emergencies on a global level. Although the exact etiology remains unknown, it has been largely linked to an obstructive cause, with subsequent elevation of pressure within the appendicular lumen leading to necrosis and rupture [1].

The presentation of the condition varies in adults and children, with adults largely presenting with right lower quadrant pain, abdominal rigidity, and a characteristic periumbilical pain radiating to the right lower quadrant, while children present with absent or diminished bowel sounds, a positive Psoas sign, positive Obturator sign, and a positive Rovsing sign [2].

Although various appendicitis scoring systems have been established to facilitate urgent patient care in patients presenting with suspected appendicitis [3], histopathological confirmation on biopsy remains the gold standard for diagnosis [4].

The Alvarado scoring system, introduced in 1986, aids in the efficient diagnosis and subsequent management of appendicitis in emergency hospitals. Largely owing to its simplicity and feasibility, the system is

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well-known [5]. It is based on eight factors, namely: right lower quadrant tenderness, leucocytosis, migration of pain, leukocyte shift to the left, temperature elevation, nausea, vomiting, anorexia and rebound pain. An Alvarado score of seven or above is essential for establishing a diagnosis of acute appendicitis [6,7].

The Tzanakis scoring system, established in 2005, takes into consideration four parameters namely, right lower quadrant tenderness, rebound tenderness, WHC count greater than 12,000/ml, and positive ultrasonography findings. A value of eight or above is required for establishing diagnosis of acute appendicitis in this system [7].

Although analyses on the specificity, sensitivity, positive and negative predictive values of the two scoring systems have been done individually, no current analysis comparing the aforementioned parameters of the two systems exists. Therefore, in this meta-analysis, we intend to compare and contrast the two systems to establish which is better in terms of an expeditious and efficient diagnosis of acute appendicitis.

Methodology

Sources and search method

Our meta-analysis was carried out in accordance with the Cochrane and PRISMA guidelines. A search of PubMed, SCOPUS and Google Scholar was conducted on the 10th of April from inception to April 2023. No filters or restrictions were set, and the following search string was used: ‘(Alvarado OR MANTRELS) AND (Tzanakis) AND (Acute Appendicitis)’.

Study selection

The results from our search were exported to a predefined spreadsheet and were manually screened to remove duplicates. Case reports and review articles were eliminated from the list. The reference lists of the included articles were also manually screened to ensure no article was missed during the initial search. The list of articles was then screened using the title and abstract to ensure relevance and thus irrelevant articles were excluded. The text of the finalised articles was then reviewed to ensure compliance with the inclusion criteria. Two assessors (A.A, R.H) separately screened the articles and any discrepancy was resolved by discussion. A PRISMA flowchart outlining the process can be found as Fig. 1.

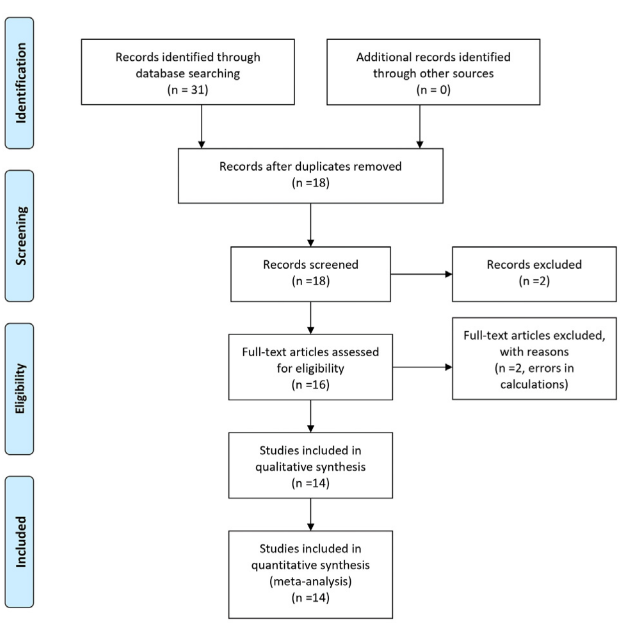

Fig. 1. PRISMA flowchart outlining the process of study selection.

The following inclusion criteria was used:

1. Patients suffering from periumbilical, right iliac fossa pain or patients suspected with acute appendicitis based on signs and symptoms,
2. Utilisation of both Alvarado and Tzanakis scores to assess risk of acute appendicitis,
3. The patients underwent appendectomy based on the results of the score,
4. The utilisation of histopathology as reference standard for confirmation of appendicitis
5. Characteristics like sensitivity, specificity, negative predictive value and positive predictive value of each score in the studies were discussed and published.

No assumptions were made regarding any missing values.

Quality assessment

The methodological quality of the included studies was assessed using the QUADAS-2 and QUADAS-C score The QUADAS-2 tool comprises of four domains with leading questions to assess the risk of bias for each of the diagnostic tests with scores defined as “low”, “unclear” or “high”. In addition to this, the QUADS-2 score also assesses concerns regarding the applicability of the score. The QUADAS-C score uses additional questions as well as data from the QUADAS-2 tool to highlight the comparative risk of bias in each of the four domains for all the included studies. Both the tools use colour coded bar graphs to accurately depict the overall risk of bias and accuracy in each of the four domains.

Statistical analysis

The Meta Disc 14 software was used for statistical analysis [8], A 95% confidence interval was set, and the software was used to calculate pooled specificity, pooled sensitivity, diagnostic OR and Summary Receiver Operator Curve (ROC) using a random effect model.

Definitions

The two scoring systems used in our analysis were the Alvarado [6] and Tzanakis [7] score, both of which use several parameters to assess the risk of acute appendicitis in a patient.

The Alvarado scoring system is based on a 10-point scoring system with the presence of elevated temperature 99.1 F, rebound tenderness, anorexia, vomiting or nausea, migration of pain to right iliac fossa and left shift (75% neutrophils) being given one point each and two points assigned to the presence of right lower quadrant tenderness and an elevated WBC count>10,000/mm, each. A cut off value of greater than seven is used as a positive score for acute appendicitis and all the included studies followed suit.

The Tzanakis score is a 15-point scoring system that includes ultrasonographic evidence to diagnose acute appendicitis. The Tzanakis score assigns four points to the presence of right lower quadrant tenderness, three points for the presence of rebound tenderness, two points for a WBC count >12,000 mm and six points for positive ultrasound findings for appendicitis. The Tzanakis score uses a value of greater than eight as diagnostic for acute appendicitis. All the included studies used the same criteria to assess a positive appendicitis rate.

Positive appendicitis was defined as histopathological confirmation of appendicitis following surgical appendectomy. This was used as the reference standard against which both the scores were compared.

Results

Included studies and baseline characteristics

Our search yielded a total of 18 studies, of which two were excluded for assessing Alvarado score alone while another two were excluded for presenting inaccurate data. Employing the inclusion criteria mentioned earlier, 14 studies [9–22] with a total of 2235 patients were included. Figure/summarises the process which resulted in the inclusion of the studies. 10 of the included studies were prospective with the rest being retrospective cohort studies. Of the 2,035 patients, 53% were male and the overall positive appendectomy rate was 83.3%, Table 1 summarises the baseline demographics and type of study. Table 2 summarises the different diagnostic values of each study.

Quality assessment

Using the QUADAS-2 tool (Fig. 2), the risk of bias for the Tzanakis score was calculated. The inclusion of patients in the studies led to a high risk of bias in 50% of the studies with the diagnostic test itself leading to a high risk of bias in only one of the fourteen studies (7%). The reference test did not contribute to a high risk of bias in any of the studies while only one study presented a high risk of bias due to the timing of administration of the tests as well as flow of the patients. On using the QUADAS-C tool (Fig. 3) to compare accuracy between the two tests, the patient inclusion caused a high risk of bias in half of the studies while both the tests themselves as well as the reference test caused a high risk of bias in around 21% of the studies. Only 5 studies of the 14 included studies had a low risk of bias in the timing of the tests and the flow of patient domain. From these results, it is evident that the greatest contributors to bias and heterogeneity in our studies were the selection of patients and the timing and flow of patients. The tests (Fig. 2) showed an

Table 1
Baseline demographics of the included studies (confirmed AA; Confirmed Acute Appendicitis on Histopathology).

Baseline Characteristics						
Author name and Year	Type of Study	Number of patients	Age	Sex		Confirmed AA
				Male	Female	
Sidgel; 2010 [9]	Prospective, non randomized, observational study	100	27.5 ± 9.8	72	28	94
Malla; 2014 [10]	Retrospective, non randomized observational study	200	N/A	N/A	N/A	184
Mithun; 2019 [11]	Prospective, non randomized cross sectional study	100	14–60 yrs.	67	33	60
Anupriya; 2019 [12]	Prospective, comparative cross sectional study	70	33.3 ± 11.08	44	26	58
Korkut; 2020 [13]	Prospective, cross sectional study	74	36.68 ± 11.97	42	32	65
Kumar; 2020 [14]	Prospective, non randomized, observational study	50	N/A	36	14	44
Sharma D; 2020 [15]	Prospective, cross sectional study	100	18–40 yrs.	54	46	88
Stiel; 2020 [16]	Retrospective, cohort study	463	10.9 ± 3.7	199	264	336
Çomçalı; 2021 [17]	Retrospective, cross sectional study	140	28.06 ± 6.42	0	140	132
Iftikhar; 2021 [18]	Prospective, cross sectional study	60	25.48 ± 9.8	34	26	51
Sharma P; 2021 [19]	Prospective, analytical study	167	28.4 ± 12.9	116	51	144
Iqbal; 2022 [20]	Cross sectional, validation study	420	20.15 ± 7.13	218	202	367
Prabahat Nichkaode; 2022 [21]	Prospective, comparative cross sectional study	150	21–40 yrs.	103	47	118
Sağ; 2022 [22]	Prospective, cross sectional study	141	11.7 ± 3.2	93	48	120

Table 2
Diagnostic values of the included studies. (PPV; Positive predictive Value, NPV; Negative Predictive Value).

Diagnostic values of Scoring Systems										
Author name and Year	Sensitivity		Specificity		PPV		NPV		Diagnostic Accuracy	
	Tzanaki	Alvarado	Tzanaki	Alvarado	Tzanaki	Alvarado	Tzanaki	Alvarado	Tzanaki	Alvarado
Sidgel; 2010 [9]	91.48%	81.91%	66.66%	66.66%	97.72%	97.46%	33.33%	19.04%	90.00%	81.00%
Malla; 2014 [10]	86.90%	76.00%	75.00%	75.00%	97.50%	97.20%	33.30%	21.40%	86.00%	76.00%
Mithun; 2019 [11]	88.24%	73.53%	93.70%	31.25%	96.77%	69.44%	78.94%	35.71%	90.00%	60.00%
Anupriya; 2019 [12]	65.52%	36.21%	100.00%	66.70%	100.00%	84.00%	37.50%	17.78%	71.43%	41.43%
Korkut; 2020 [13]	84.40%	60.90%	99.88%	89.90%	98.25%	97.56%	47.06%	24.24%	86.49%	64.86%
Kumar; 2020 [14]	79.62%	59.09%	83.30%	33.30%	97.20%	86.60%	35.70%	10.00%	80.00%	56.00%
Sharma D; 2020 [15]	80.60%	11.30%	100.00%	100.00%	100.00%	100.00%	41.30%	13.30%	83.00%	22.00%
Stiel; 2020 [16]	80.40%	67.90%	87.40%	78.70%	94.40%	89.40%	62.70%	48.10%	82.30%	70.84%
Çomçalı; 2021 [17]	90.91%	70.45%	87.50%	75.00%	99.17%	97.89%	36.84%	13.33%	90.71%	70.71%
Iftikhar; 2021 [18]	94.11%	74.00%	88.88%	55.00%	97.95%	90.00%	72.72%	27.00%	93.33%	71.66%
Sharma P; 2021 [19]	93.80%	56.90%	30.40%	91.30%	89.40%	97.60%	43.80%	25.30%	85.00%	61.70%
Iqbal; 2022 [20]	88.83%	86.92%	88.68%	92.45%	98.19%	98.76%	53.41%	50.52%	88.81%	86.62%
Prabahat Nichkaode; 2022 [21]	72.30%	27.73%	3.20%	80.65%	74.10%	84.62%	2.90%	22.52%	58.00%	38.67%
Sağ; 2022 [22]	97.50%	80.80%	14.30%	42.90%	86.70%	88.90%	50.00%	28.10%	85.10%	75.20%

Fig. 2. QUADAS-2 showing the risk of bias for Tzanakis score.

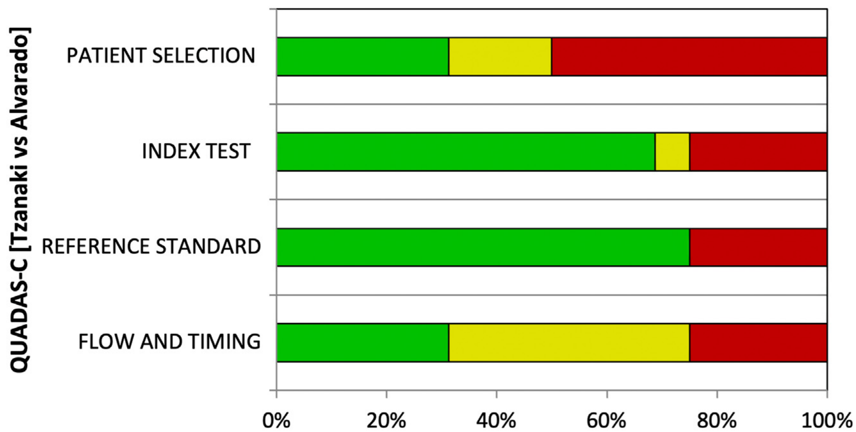

Fig. 3. QUADAS-C showing the comparative risk of bias for Alvarado and Tzanakis score.

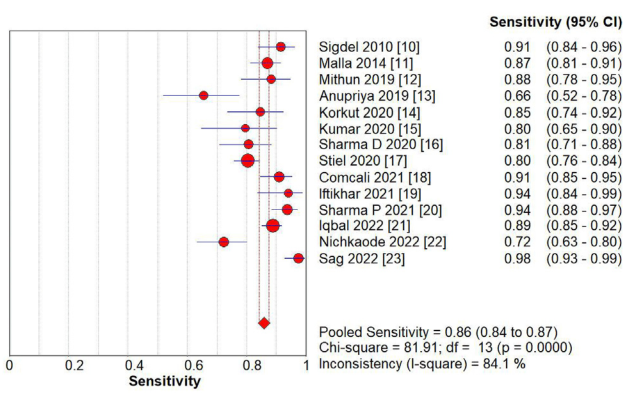

Fig. 4. Pooled sensitivity of Tzanakis score.

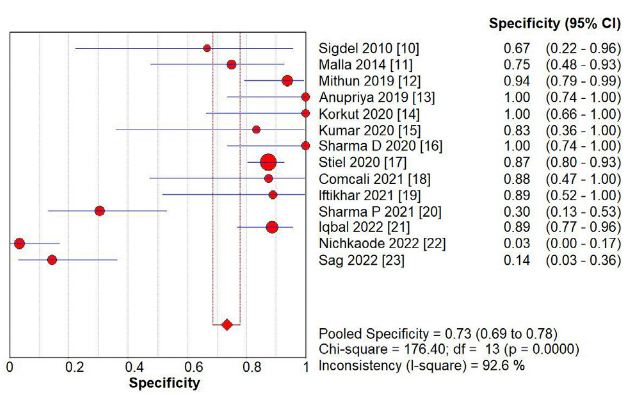

Fig. 5. Pooled specificity of Tzanakis score.

exceptional 100% low concern in applicability for both patient standard and reference standard and an 79% low rate of concern in the index test.

Statistical outcomes

Our results show that the pooled sensitivity of Tzanakis score (Fig. 4) was 0.86 (95% CI; 0.84–0.87) while the pooled specificity (Fig. 5) was 0.73 (95% CI; 0.69–0.78). The pooled diagnostic odds ratio (OR) for Tzanakis score (Fig. 6) was 22.52 (95% CI; 9.47–53.56) while the AUC for Tzanakis score (Fig. 7) was 0.9261 (SE; 0.0169).

As for the Alvarado score, our analysis showed a pooled sensitivity (Fig. 8) of 0.67 (95% CI; 0.65–0.69) and a pooled specificity (Fig. 9) of 0.74 (95% CI; 0.69–0.79). The diagnostic odds ratio (OR) (Fig. 10) was 4.92 (95% CI; 2.48–9.75) and the area under the curve (AUC) (Fig. 11) was 0.7389 (SE; 0.0489).

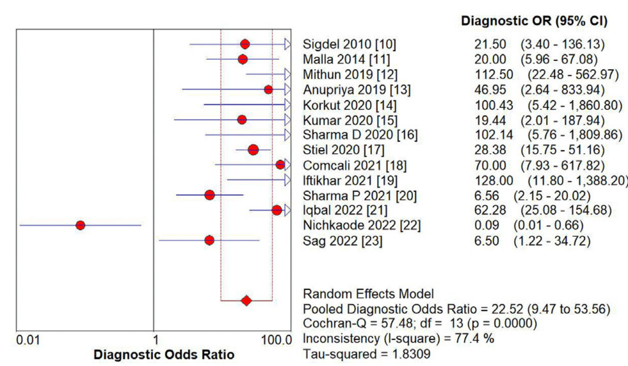

Fig. 6. Pooled diagnostic odds ratio of Tzanakis score.

Discussion

The results of our analysis indicate that the Tzanakis score is superior to the Alvarado scoring system in terms of sensitivity hence making it more accurate in diagnosing appendicitis positively. However, the Alvarado score has a marginally better specificity indicating its superiority over the Tzanakis score in ruling out acute appendicitis. Nevertheless, the diagnostic odds ratio (OR) and area under the curve (AUC) for the Tzanakis score was much greater than the Alvarado score proving it as the better scoring system of the two.

A previous meta-analysis comparing the Alvarado scoring system to the RIPASA scoring system has been published by Maximos Frountzas et al. [23] However, to our knowledge, this is the first meta-analysis comparing the Alvarado and Tzanakis score in terms of pooled specificity, sensitivity, diagnostic odds ratio and SROC curves.

Although imaging techniques such as computed tomography facilitate the diagnostic process, the adverse effects associated with radiation exposure have demanded and led to a re-emphasis on the establishment of a clinical diagnosis for the condition [24]. In addition to this, access to computed tomography in emergency settings is limited in resource poor countries as well as rural areas across the globe [25], necessitating the need for clinical diagnosis to avoid negative appendectomies or the unnecessary transfer of patients to tertiary care centres for evaluation of appendicitis. Unlike computed tomography, ultrasonography is relatively more readily available across emergency settings aiding in a more accurate diagnosis in the Tzanakis scoring system.

Additionally, it is difficult to establish a diagnosis of appendicitis in the elderly, children under three, and pregnant patients [26]. It has been

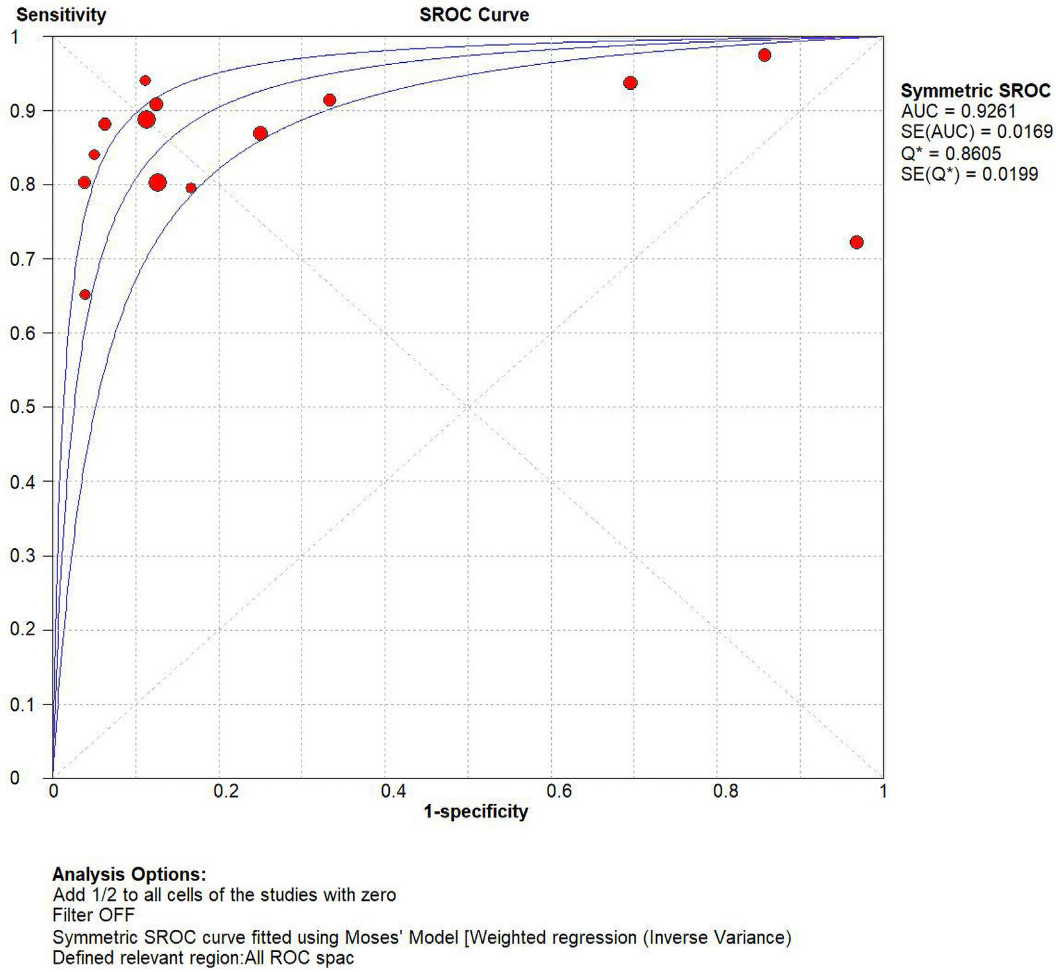

Fig. 7. SROC curve for Tzanakis score.

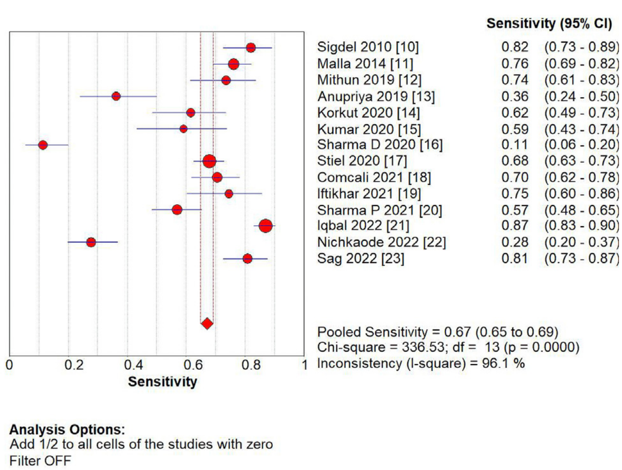

Fig. 8. Pooled sensitivity of Alvarado score.

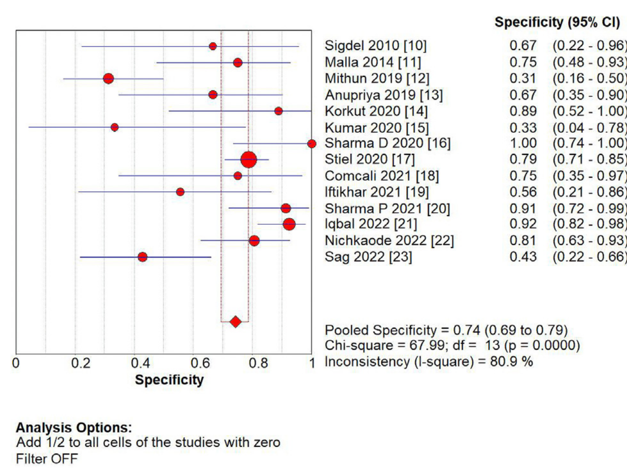

Fig. 9. Pooled specificity of Alvarado score.

reported that one out of every five cases of appendicitis are misdiagnosed and about 15–35% of appendices removed in emergency appendectomy are normal on histopathological findings [27]. In our analysis, the combined negative appendectomy rate was 16.7%. It is, therefore, imperative to consider ways of establishing a clinical diagnosis, eliminating the need for unnecessary surgery.

The main advantage of this meta-analysis is that it provides an evidential and statistical basis of the efficiency of a scoring system that is widely applicable in clinical settings, easy to use and decipher, cost-effective and proficient for a condition that is commonly encountered in surgical emergencies of all countries.

Our analysis indicates that the Tzanakis score is superior to Alvarado score in terms of sensitivity, as well as diagnostic accuracy. This is most likely due to the use of ultrasonography in positively identifying patients with acute appendicitis. The Alvarado score uses clinical findings as well as laboratory evidence to support the diagnosis of acute appendicitis.

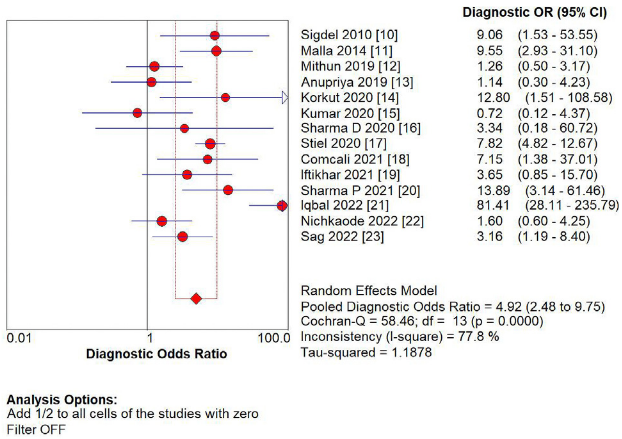

Fig. 10. Pooled diagnostic odds ratio of Alvarado score.

However, Tzanakis system adds a third component in the form of imaging to provide a higher accuracy in diagnosing acute appendicitis.

Nevertheless, the use of Tzanakis system in pregnant patients is limited when ultrasonography findings are inconclusive owing to the gravid uterus, especially in the third trimester of pregnancy [28]. Ultrasonographic limitations also apply to obese patients and cases of atypical appendicitis, in which CT scans have been recommended [29].

It should also be noted that the Tzanakis scoring system takes into consideration ultrasonographic findings, the results of which are both operator and instrument dependent and prone to variation. Resultantly, the specificity of Tzanakis system varies accordingly. For this very reason, it would likely be more time consuming as a scoring system, when compared to Alvarado score, which is established based on patient symptomatology and clinical signs only. However, no existing literature could be found to support the postulation quantitatively. The use of Tzanakis system would also involve the cost of ultrasonographic findings, making it relatively less cost-effective. It is imperative to note, however, that the Alvarado system depends largely on physician experience to successfully elicit and identify the clinical signs necessary for making an accurate diagnosis. These signs can easily be overlooked or missed altogether by an inexperienced or overworked surgeon. This is particularly a problem in underdeveloped and rural areas with fewer specialist doctors and an increased patient load often causing burnout in surgeons.

Limitations

Our analysis has several limitations. Firstly, the individual studies analysed had small sample sizes, with the smallest sample size being 50 and the largest sample being 463. Secondly, owing to the use and easy access of imaging techniques such as CT scans and high-resolution ultrasound in Western health settings, most of the studies were conducted in non-Western health systems with considerate bias in the selected studies. Thirdly, this study does not compare the individual scoring systems with imaging modalities such as a CT scan and ultrasound findings. It is also pertinent to note that both the Alvarado and Tzanakis systems make use of clinical signs such as tenderness and rebound tenderness which can be better elicited and appreciated by clinicians

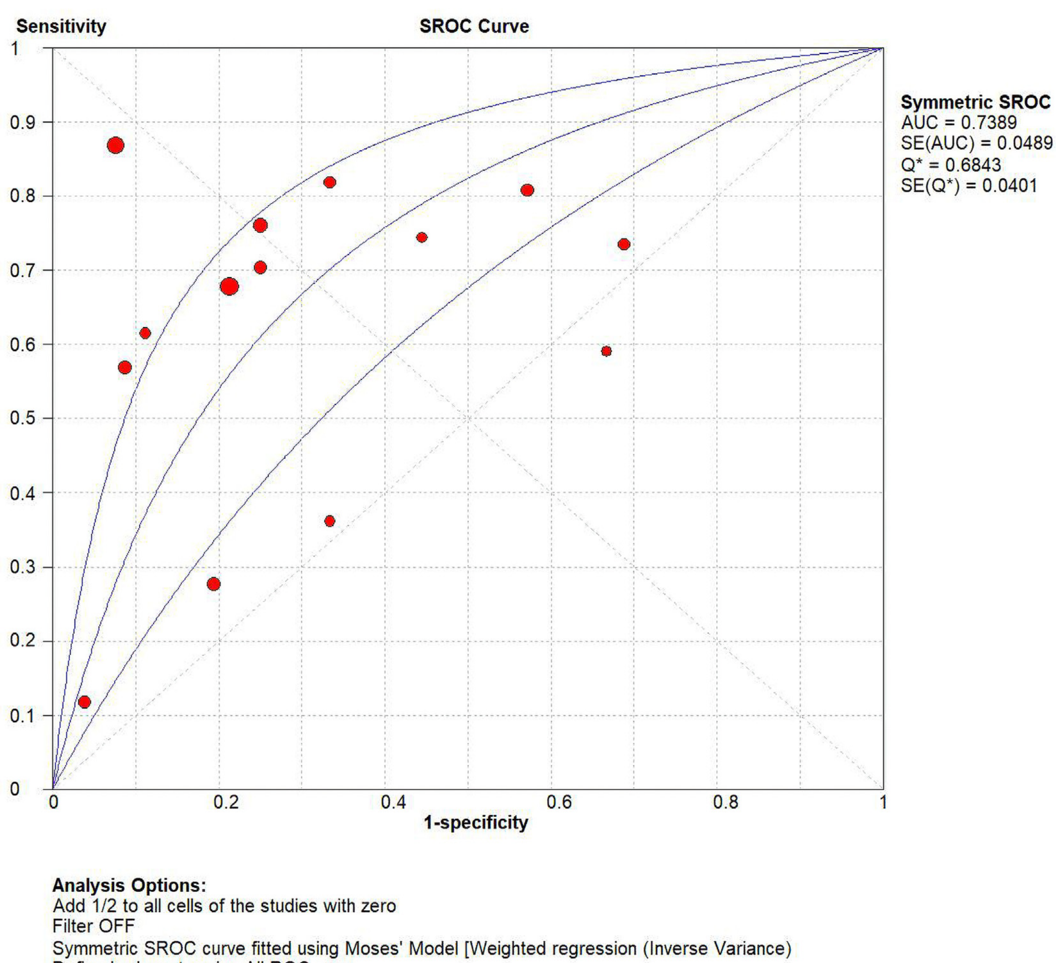

Fig. 11. SROC curve for Alvarado score.

with adequate experience leading to a risk of bias if elicited by inexperienced surgeons.

In addition to this, patient selection, timing and flow were sources of significant bias in the individual studies on quality assessment, as has been stated earlier.

Conclusions

The Tzanakis scoring system is more sensitive as well as more accurate in making a diagnosis of acute appendicitis when compared to the Alvarado scoring system, making it more useful in the timely and accurate diagnosis of appendicitis. On the other hand, the Alvarado scoring system has a slightly higher specificity when compared to the Tzanakis scoring system. However, it is pertinent to note that the Tzanakis scoring system relies heavily on the use of ultrasonography for diagnosis, with problems arising due to either a lack of availability of resource or a lack of trained individuals to make an accurate observation on ultrasound. Based on our analysis, it is safe to assume that the Tzanakis scoring system can replace the Alvarado scoring system in hospitals where emergent ultrasonography is available. This will ensure a reduction in negative appendectomy rates and thus reduce burden on both the hospital as well as the surgeons.

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Authors' contributions

1. Abdul Rafeh Awan: literature review, data screening, data extraction, data analysis, manuscript writing, manuscript compilation.
2. Zaka Ullah Khan: idea formulation, literature review, data screening, data extraction.
3. Hira Saleem: data screening, data extraction, manuscript writing, manuscript compilation.
4. Hanniya Iqbal: data screening, data extraction, manuscript writing.
5. Waqas Ahmad: data screening, data extraction, manuscript writing.
6. Ali Raza Khan: data screening, data extraction.
7. Mobeen Farooqi: manuscript editing, reviewing and publication.

Declaration of Generative AI and AI-assisted technologies in the writing process

Authors declare that A.I was not used in the process of this manuscript.

Declaration of competing interest

All authors declare no conflict of interest.

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